Hints: Limits – 9. Sequences defined by integrals and sums

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. The Riemann sum $\frac{1}{n}\sum f(\frac{k}{n})$ lies between the lower and upper Darboux sums of the uniform partition, and both tend to $\int_0^1 f$ for integrable f. For monotone (say increasing) f, on each piece $\int_{(k-1)/n}^{k/n} f$ lies between $\frac{1}{n}f(\frac{k-1}{n})$ and $\frac{1}{n}f(\frac{k}{n})$. Summing, the error is at most $\frac{1}{n}\sum_k (f(\frac{k}{n}) - f(\frac{k-1}{n})) = \frac{f(1)-f(0)}{n}$.

Hint 2. For smooth f, Taylor-expand around the right endpoint: $\frac{1}{n}f(\frac{k}{n}) - \int_{(k-1)/n}^{k/n} f = \frac{1}{2n^2}f'(\frac{k}{n}) + O(n^{-3})$. Summing, $\frac{1}{2n}\cdot\frac{1}{n}\sum f'(\frac{k}{n}) + O(n^{-2}) = \frac{f(1)-f(0)}{2n} + O(n^{-2})$.

Hint 3. The two sums in the question are the same sum, because $\frac{1}{n}\cdot\frac{1}{1+(k/n)^2} = \frac{n}{n^2+k^2}$. Both limits equal $\int_0^1 \frac{dx}{1+x^2} = \frac{\pi}{4}$.

Problem 2

Hint 1. First take $f = \mathbf{1}_{[c,d]}$. Then $\int_c^d g(nx)\,dx = \frac{1}{n}\int_{nc}^{nd} g$. The interval $[nc, nd]$ contains about $n(d-c)/T$ full periods, and the incomplete pieces contribute $O(\frac{1}{n})$. So the integral tends to $(d-c)\cdot\frac{1}{T}\int_0^T g$. By linearity the same holds for step functions.

Hint 2. For integrable f and $\varepsilon > 0$, choose a step function s with $\int_a^b |f-s| < \varepsilon$. Since g is bounded, $\left|\int (f-s)g(nx)\right| \leqslant \sup|g|\cdot\varepsilon$. (A Riemann integrable g is bounded.) For Riemann–Lebesgue take $g = \sin$, whose mean is 0.

Hint 3. In the example, $g(y) = \frac{1}{1+3\cos^2 y}$ has period π, and $\frac{1}{\pi}\int_0^\pi \frac{dy}{4\cos^2 y + \sin^2 y} = \frac{1}{\pi}\cdot\frac{\pi}{2\cdot 1} = \frac{1}{2}$ (substitute $u = \tan y$). Replacing n by $2n$ changes nothing. The limit is $\frac{1}{2}\int_0^\pi \sin x\,dx = 1$.

Problem 3

Hint 1. The integrand is largest near $x = 1$. Compare a_n with $\int_0^1 x^n\,dx$, and use $a_n + a_{n+1} = \int_0^1 x^n\,dx = \frac{1}{n+1}$.

Hint 2. (a_n) decreases, so $2a_{n+1} \leqslant a_n + a_{n+1} = \frac{1}{n+1} \leqslant 2a_n$. Hence $a_n \sim \frac{1}{2n}$. The series $\sum(-1)^n a_n$ converges by Leibniz's test.

Hint 3. To get the sum, sum the geometric series under the integral: $\sum_{n=1}^{N}(-1)^n x^n = \frac{-x+(-x)^{N+1}}{1+x}$. The remainder term integrates to at most $\int_0^1 x^{N+1}dx \to 0$. So the sum is $-\int_0^1 \frac{x}{(1+x)^2}dx = \frac{1}{2} - \ln 2$.

Problem 4

Note. As stated, the limit is not determined by C. If $C < 1$, both series converge and the ratio of the sums depends on f; for example, change f on $[0,1]$ only, which changes b_1 alone. If $C > 1$, take $f(t) = C\exp(\varphi(t)/\sqrt{t})$ with $\varphi \approx 1$ near m^2 for odd m and $\varphi \approx 0$ near m^2 for even m. Then $f \to C$, but the ratio oscillates between $\frac{eC+1}{C+e}$ and its reciprocal. (For $f \equiv C$ the ratio is exactly 1.)

The hints establish the key relation and the correct general statement: if $A_n = a_1 + \cdots + a_n \to \infty$ and $a_{n+1}/A_n \to L$, the limit is $\frac{C}{1+L}$.

Hint 1. Look at which interval b_{n+1} uses. It is $[n^2, n^2+1]$, the same interval as a_n, but with exponent $n+1$ instead of n. Compare b_{n+1} with a_n.

Hint 2. $b_{n+1} = \int_{n^2}^{n^2+1} f \cdot f^n$, so $\min_{[n^2,n^2+1]} f \leqslant b_{n+1}/a_n \leqslant \max_{[n^2,n^2+1]} f$, and both bounds tend to C. Hence $b_{n+1} = (C + o(1))a_n$. (Note $a_n > 0$ for large n, since $f > 0$ far out.)

Hint 3. Therefore $b_1 + \cdots + b_{n+1} = b_1 + \sum_{k \leqslant n}(C + o(1))a_k = (C + o(1))A_n$ when $A_n \to \infty$ (a Cesàro/Stolz argument). The denominator is $A_{n+1} = A_n(1 + a_{n+1}/A_n)$. So the ratio tends to $\frac{C}{1+L}$ whenever $a_{n+1}/A_n \to L$. For instance, when $a_{n+1}/A_n \to 0$ the limit is C, and for $f \equiv C > 1$ one has $L = C - 1$ and the limit is 1.

Problem 5

Hint 1. Substitute $x = \frac{\pi}{2} - t$: the integral becomes $\int_0^{\pi/2} \cos^n t \, dt$, and the mass concentrates near $t = 0$, where $\cos^n t = e^{n \ln \cos t} \approx e^{-nt^2/2}$.

Hint 2. Split at $t = n^{-1/3}$ (so that $nt^3 \to 0$ but $nt^2 \to \infty$). On $[0, n^{-1/3}]$, $n \ln \cos t = -\frac{nt^2}{2}(1 + O(t^2))$, and after $t = s/\sqrt{n}$ the integral is $\frac{1}{\sqrt{n}}(\int_0^\infty e^{-s^2/2} ds + o(1)) = \frac{1}{\sqrt{n}}\sqrt{\frac{\pi}{2}}\,(1 + o(1))$. The rest is at most $\frac{\pi}{2}\cos^n(n^{-1/3}) \leqslant \frac{\pi}{2} e^{-cn^{1/3}}$.

Hint 3. Wallis: the exact recursion gives $a_{2m} = \frac{(2m-1)!!}{(2m)!!}\frac{\pi}{2}$ and $a_{2m+1} = \frac{(2m)!!}{(2m+1)!!}$. The asymptotics give $a_{2m}/a_{2m+1} \to 1$, which is Wallis' product $\frac{\pi}{2} = \prod_{k \geqslant 1} \frac{(2k)^2}{(2k-1)(2k+1)}$. With the gamma function, $a_n = \frac{\sqrt{\pi}}{2}\frac{\Gamma(\frac{n+1}{2})}{\Gamma(\frac{n}{2}+1)}$ (a Beta integral), and $\Gamma(x + \frac{1}{2})/\Gamma(x+1) \sim x^{-1/2}$ (from Stirling) gives the same $\sqrt{\pi/(2n)}$.